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Research Data Scientist with PhD in Mathematics, specializing in advanced neural networks. Over **9 years of experience** transforming complex mathematical concepts into practical, high-impact solutions - with results deployed at enterprise level for clients such as

Boeing, Lockheed Martin, and British Petroleum.

- Proficient in **SQL, Power BI, and Python**, with expertise in data visualization, statistical and mathematical modelling, and machine learning libraries such as **PyTorch** and **LibTorch**.
- Proven ability to perform **in-depth analyses, forecast customer behavior**, and optimize **algorithms** to drive business outcomes.
- Holds Ph.D. degree in Mathematics from the University of Texas at Dallas.

KEY ACHIEVEMENTS

- Drove loyalty program **ROI** up by 23% through Python-based partner selection model that optimized point allocation strategy while maintaining 85% prediction accuracy.
- Built an **SQL**-based program to automate the generation of marketing segments, reducing the selection time from 30 minutes to under 1 minute and eliminating the dependency of the **CRM** team.
- Cut composite material testing costs by \$50K annually with **Physically Recurrent Neural Networks** (PRNN) implementation that delivered 100x processing acceleration and required 10x less training data, enabling faster product development cycles.

TECHNICAL SKILLS

Quantitative Analysis: Data Analysis, Visualization, Linear regression, Monte Carlo

Machine Learning: Predictive Modeling, Gradient Boosting, Random Forests

Programming and Software: Python (PyTorch, Numpy, Pandas, Matplotlib), C++ (LibTorch)

DBMS: SQL, Oracle, Power BI

Misc.: Mathematica, Matlab, LATEX, Excel, Power Point, GitLab

OS: Windows, Linux

PROFESSIONAL EXPERIENCE

UNIVERSITY OF TEXAS AT ARLINGTON RESEARCH INSTITUTE (UTARI)

August 2024 – present

UTARI specializes in applying cutting-edge technologies to real-world engineering problems.

Postdoctoral research associate - ML Data Analyst

Led ML-driven research in computational mechanics for Boeing and Lockheed, focusing on PRNN models.

- Reduced composite material development cycles by 5%, saving ~\$50K per project through accurate stress-strain forecasting with PRNN.
- Developed and implemented data-driven models to simulate material responses, doubling design efficiency.
- Presented results at two international conferences, enhancing institutional and industry visibility.
- Reduced composite component testing cycle from 7 months to 4 months through advanced simulation models.

Technologies and Tools:

- Proficient in **Python** (Numpy, Pandas, PyTorch, Matplotlib), **C++** (LibTorch), **MATLAB**, and **Mathematica** for numerical modeling and data visualization.
- Experience with **high-performance computing** (HPC) environments and version control systems (**Git**).

UNIVERSITY OF TEXAS AT DALLAS (UTD)

August 2018 – December 2023

Top 65 U.S. Statistics Program, known for applied data science and analytics.

Research assistant/Teaching assistant

Leading research university with strong focus on computational sciences and mathematics.

- Led research team of 7 students in developing novel computational methods for dynamical systems.
- Leveraged **machine learning** to uncover long-term statistical behavior in complex systems, advancing research in ergodic theory.

LOYALTY PARTNERS VOSTOK (LPV)

June 2015 – July 2018

LPV provided outsourced loyalty marketing for major retailers (e.g. BP) and B2C brands with 5+ million active users.

Data Analyst -> Leading analyst and BI specialist

Data Analysis and Insights

- Created detailed customer profiles through deep-dive analytics, enabling highly targeted campaigns and increasing marketing effectiveness by 30%.
- Forecasted **long-term customer behavior** using Python-based **statistical models**, contributing to loyalty strategies that helped retain **thousands of high-value customers**.
- Assessed the impact of marketing programs using **SQL** and **Excel**; insights drove partner outreach efforts, resulting in the addition of **15+ new partner brands**, including top-tier retailers.

Business Intelligence

- Partnered with the **DB** team to improve **BI system**; launched automated **Power BI** reports, reducing reporting time by 80%.
- Built an **SQL** tool to auto-generate communication segments, achieving a 30x speedup and reducing analyst workload.

Identified data-modeling limitations that inspired further academic research, leading to pursuit of doctoral studies in mathematics.

EDUCATION

University of Texas at Dallas (UTD)

Ph.D. in Mathematics, Thesis title: Ergodic properties of generalized billiards.

Lomonosov Moscow State University (MSU)

Ph.D. in Mathematics, Thesis title: Topology of ellipsoid of revolution on the smooth plane

Lomonosov Moscow State University (MSU)

B.Sc./M.Sc. in Mechanics (Applied Mathematics)

AWARDS, GRANTS AND PUBLICATIONS

NSF Finite Dimensional Integrable Systems	2022
RFBR Geometry and Topology	2019
Leading Scientific Schools	2018
Multiple mathematics competition winner	2010-2012
<i>Papers:</i>	
Ergodic properties of generalized billiards	2024
Topology of dynamics of a nonhomogeneous rotationally symmetric ellipsoid on a smooth plane	2018
Minimal triangulations of two-dimensional manifolds	2016

PROFESSIONAL DEVELOPMENT

Oracle Cloud Infrastructure Foundations (Coursera)	May 2022
Python (Stepik)	January 2019
HSBC Global Banking & Markets Job Simulation BI/Data analyst (Forage)	June 2024
Accenture North America Data Analytics and Visualization Job Simulation Business analyst/Data analyst (Forage)	June 2024
Algorithms in mL (Stepik)	September 2024
LeetCode Problem Solving	Ongoing
AWS Skill Builder (Amazon)	Ongoing

EXECUTIVE STRENGTHS

MATHEMATICAL FOUNDATIONS:	Advanced Statistics, Dynamical Systems, Ergodic Theory, Topology
MACHINE LEARNING:	Physics-Informed Neural Networks, Deep Learning Architectures, Gradient Boosting, Reinforcement Learning
DATA ENGINEERING:	ETL Pipeline Design, Database Optimization, Big Data Processing, Cloud Infrastructure
LANGUAGES & TOOLS:	Python (Advanced), C++ (Advanced), SQL (Advanced), Excel, MATLAB, Mathematica, LaTeX
FRAMEWORKS:	PyTorch, LibTorch, NumPy, Pandas
SPECIALIZED:	High-Performance Computing, Parallel Processing, Mathematical Modeling, Scientific Computing